

1. Solving a System of Linear Equations

Use Python to solve the following system of equations. **Hint:** You can write this system as a matrix equation in the form $\mathbf{Ax} = \mathbf{b}$, where the matrix $\mathbf{A}$ is invertible.

$$\begin{array}{ccccccccc} 27x_1 & - & 10x_2 & + & 4x_3 & - & 29x_4 & = & 1 \\ -16x_1 & + & 5x_2 & - & 2x_3 & + & 18x_4 & = & -1 \\ -17x_1 & + & 4x_2 & - & 2x_3 & + & 20x_4 & = & 0 \\ -7x_1 & + & 2x_2 & - & x_3 & + & 8x_4 & = & 1 \end{array}$$

2. Visualizing a Linear Transformation

Let T be the linear transformation that takes in a vector $\mathbf{v}$ in $\mathbb{R}^2$ and outputs the product of the 2×2 matrix $\mathbf{A}$ shown below and $\mathbf{v}$.

$$\mathbf{A} = \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$$

I.e., T is a function that takes in a vector $\mathbf{v}$ in $\mathbb{R}^2$ and outputs the vector $T(\mathbf{v}) = \mathbf{Av}$, also in $\mathbb{R}^2$.

- (a) Find $T(\mathbf{x})$, where $\mathbf{x} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$.
- (b) Mimic the code in Section 7.1.3 of *Essential Math for Data Science* to visualize the linear transformation T .
- (c) Use your visualization from (2) to describe geometrically how T transforms vectors in general.

3. Finding the Eigenvalues and Eigenvectors of a Matrix

Let $\mathbf{A}$ be the matrix given below.

$$\mathbf{A} = \begin{bmatrix} 4 & 0 & 1 \\ -1 & -6 & -2 \\ 5 & 0 & 0 \end{bmatrix}$$

- (a) Is the vector $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ an eigenvector of $\mathbf{A}$? Verify your answer with a calculation in Python.
- (b) Is the vector $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ an eigenvector of $\mathbf{A}$? Again, verify your answer with a calculation in Python.
- (c) Use Python to find all of the eigenvalues and eigenvectors of $\mathbf{A}$.

4. Principal Component Analysis

For this exercise, you will be working with the data set **video_game_data.csv**. This data has four columns: name of the game, critic score, user score, and total sales. **Note:** This data set is a modified version of the data set from <https://www.kaggle.com/ashaheedq/video-games-sales-2019>.

- (a) Make a scatterplot of the user scores versus critics scores.
- (b) On your scatterplot from part (a), sketch the approximate directions of the first and second principal components. This can be done by hand or in Python.